

Results

Descriptive Statistics

Descriptive Statistics

	Age	Gender	Level of Study	Academic Program	Year of Study	Blockchain Knowledge	Understanding of IoT
Valid	100	100	100	100	100	100	100
Missing	0	0	0	0	0	0	0
Mean							
Std. Deviation							
Minimum							
Maximum							

Note. Not all values are available for *Nominal Text* variables

Frequency Tables

Frequencies for Age

Age	Frequency	Percent	Valid Percent	Cumulative Percent
18-21	14	14.000	14.000	14.000
22-25	59	59.000	59.000	73.000
26?30	19	19.000	19.000	92.000
Above 30	8	8.000	8.000	100.000
Missing	0	0.000		
Total	100	100.000		

Frequencies for Gender

Gender	Frequency	Percent	Valid Percent	Cumulative Percent
Female	31	31.000	31.000	31.000
Male	62	62.000	62.000	93.000
Other	3	3.000	3.000	96.000
Prefer not to say	4	4.000	4.000	100.000
Missing	0	0.000		
Total	100	100.000		

Frequencies for Level of Study

Level of Study	Frequency	Percent	Valid Percent	Cumulative Percent
Other	6	6.000	6.000	6.000
Postgraduate (Master?s)	21	21.000	21.000	27.000
Undergraduate	73	73.000	73.000	100.000
Missing	0	0.000		
Total	100	100.000		

Frequencies for Academic Program

Academic Program	Frequency	Percent	Valid Percent	Cumulative Percent
Business Administration	12	12.000	12.000	12.000
Computer Science / IT	52	52.000	52.000	64.000
Economics	11	11.000	11.000	75.000
Other	13	13.000	13.000	88.000
Social Sciences	12	12.000	12.000	100.000
Missing	0	0.000		
Total	100	100.000		

Frequencies for Year of Study

Year of Study	Frequency	Percent	Valid Percent	Cumulative Percent
1st Year	8	8.000	8.000	8.000
2nd Year	66	66.000	66.000	74.000
3rd Year	19	19.000	19.000	93.000
4th Year or above	7	7.000	7.000	100.000
Missing	0	0.000		
Total	100	100.000		

Frequencies for Blockchain Knowledge

Blockchain Knowledge	Frequency	Percent	Valid Percent	Cumulative Percent
No	22	22.000	22.000	22.000
Yes	78	78.000	78.000	100.000
Missing	0	0.000		
Total	100	100.000		

Frequencies for Understanding of IoT

Understanding of IoT	Frequency	Percent	Valid Percent	Cumulative Percent
Fair	24	24.000	24.000	24.000
Good	48	48.000	48.000	72.000
Poor	6	6.000	6.000	78.000
Very Good	18	18.000	18.000	96.000
Very Poor	4	4.000	4.000	100.000
Missing	0	0.000		
Total	100	100.000		

Unidimensional Reliability for Student Awareness

Frequentist Scale Reliability Statistics

Estimate	Cronbach's α
Point estimate	0.964
95% CI lower bound	0.951
95% CI upper bound	0.974

Frequentist Individual Item Reliability Statistics

Item	If item dropped
	Cronbach's α
SA1	0.948
SA2	0.958
SA3	0.956
SA4	0.958
SA5	0.957

Unidimensional Reliability for Perceived Ease of Use

Frequentist Scale Reliability Statistics

Estimate	Cronbach's α
Point estimate	0.963
95% CI lower bound	0.950
95% CI upper bound	0.973

Frequentist Individual Item Reliability Statistics

Item	If item dropped
	Cronbach's α
PEU1	0.947
PEU2	0.956
PEU3	0.955
PEU4	0.956
PEU5	0.956

Unidimensional Reliability for Perceived Security Benefits

Frequentist Scale Reliability Statistics

Estimate	Cronbach's α
Point estimate	0.944
95% CI lower bound	0.924
95% CI upper bound	0.959

Frequentist Individual Item Reliability Statistics

Item	If item dropped
	Cronbach's α
PSB1	0.933
PSB2	0.933
PSB3	0.931
PSB4	0.926
PSB5	0.929

Unidimensional Reliability for Perceived Usefulness

Frequentist Scale Reliability Statistics

Estimate	Cronbach's α
Point estimate	0.947
95% CI lower bound	0.928
95% CI upper bound	0.962

Frequentist Individual Item Reliability Statistics

Item	If item dropped
	Cronbach's α
PU1	0.931
PU2	0.932
PU3	0.932
PU4	0.937
PU5	0.942

Unidimensional Reliability for Blockchain Acceptance

Frequentist Scale Reliability Statistics

Estimate	Cronbach's α
Point estimate	0.936
95% CI lower bound	0.913
95% CI upper bound	0.954

Frequentist Individual Item Reliability Statistics

Item	If item dropped
	Cronbach's α
ACC1	0.916
ACC2	0.920
ACC3	0.924
ACC4	0.922
ACC5	0.923

Confirmatory Factor Analysis for Student Awareness

Model fit

Chi-square test

Model	X ²	df	p
Baseline model	621.293	10	
Factor model	3.343	5	0.647

Note. The estimator is ML.

Kaiser-Meyer-Olkin (KMO) test

Indicator	MSA
SA1	0.853
SA2	0.931
SA3	0.914
SA4	0.938
SA5	0.939
Overall	0.912

Bartlett's test of sphericity

X ²	df	p
599.547	10	< .001

R-Squared

	R ²
SA1	0.947
SA2	0.809
SA3	0.847
SA4	0.812
SA5	0.807

Parameter estimates

Factor loadings

Factor	Indicator	Estimate	Std. Error	z-value	p	95% Confidence Interval	
						Lower	Upper
Student Awareness	SA1	1.346	0.101	13.355	< .001	1.149	1.544
	SA2	1.278	0.110	11.590	< .001	1.062	1.495
	SA3	1.379	0.114	12.062	< .001	1.155	1.603
	SA4	1.239	0.107	11.632	< .001	1.030	1.448
	SA5	1.264	0.109	11.561	< .001	1.050	1.478

Factor variances

Factor	Estimate	Std. Error	z-value	p	95% Confidence Interval	
					Lower	Upper
Student Awareness	1.000	0.000			1.000	1.000

Residual variances

Indicator	Estimate	Std. Error	z-value	p	95% Confidence Interval	
					Lower	Upper
SA1	0.101	0.029	3.419	< .001	0.043	0.159
SA2	0.385	0.063	6.161	< .001	0.263	0.508
SA3	0.343	0.058	5.947	< .001	0.230	0.457
SA4	0.355	0.057	6.183	< .001	0.242	0.467
SA5	0.382	0.062	6.135	< .001	0.260	0.504

Average variance extracted

Factor	AVE
Student Awareness	0.844

Confirmatory Factor Analysis for Perceived Ease of Use

Model fit

Chi-square test

Model	X ²	df	p
Baseline model	605.490	10	
Factor model	4.088	5	0.537

Note. The estimator is ML.

Kaiser-Meyer-Olkin (KMO) test

Indicator	MSA
PEU1	0.861
PEU2	0.930
PEU3	0.913
PEU4	0.936
PEU5	0.933
Overall	0.913

Bartlett's test of sphericity

X ²	df	p
584.297	10	< .001

R-Squared

	R ²
PEU1	0.935
PEU2	0.803
PEU3	0.842
PEU4	0.811
PEU5	0.808

Parameter estimates

Factor loadings

Factor	Indicator	Estimate	Std. Error	z-value	p	95% Confidence Interval	
						Lower	Upper
Factor 1	PEU1	1.320	0.100	13.185	< .001	1.123	1.516
	PEU2	1.260	0.110	11.498	< .001	1.045	1.475
	PEU3	1.344	0.112	11.989	< .001	1.124	1.564
	PEU4	1.232	0.106	11.608	< .001	1.024	1.440
	PEU5	1.255	0.109	11.558	< .001	1.042	1.468

Factor variances

Factor	Estimate	Std. Error	z-value	p	95% Confidence Interval	
					Lower	Upper
Factor 1	1.000	0.000			1.000	1.000

Residual variances

Indicator	Estimate	Std. Error	z-value	p	95% Confidence Interval	
					Lower	Upper
PEU1	0.121	0.032	3.770	< .001	0.058	0.183
PEU2	0.390	0.064	6.093	< .001	0.265	0.516
PEU3	0.339	0.058	5.875	< .001	0.226	0.452
PEU4	0.353	0.058	6.095	< .001	0.240	0.467
PEU5	0.375	0.062	6.032	< .001	0.253	0.496

Average variance extracted

Factor	AVE
Factor 1	0.839

Confirmatory Factor Analysis for Perceived Security Benefits

Model fit

Chi-square test

Model	X ²	df	p
Baseline model	461.250	10	
Factor model	11.318	5	0.045

Note. The estimator is ML.

Kaiser-Meyer-Olkin (KMO) test

Indicator	MSA
PSB1	0.917
PSB2	0.874
PSB3	0.886
PSB4	0.886
PSB5	0.892
Overall	0.891

Bartlett's test of sphericity

X ²	df	p
445.106	10	< .001

R-Squared

	R ²
PSB1	0.743
PSB2	0.752
PSB3	0.763
PSB4	0.814
PSB5	0.781

Parameter estimates

Factor loadings

Factor	Indicator	Estimate	Std. Error	z-value	p	95% Confidence Interval	
						Lower	Upper
Perceived Security Benefits	PSB1	1.233	0.115	10.698	< .001	1.007	1.459
	PSB2	1.238	0.115	10.785	< .001	1.013	1.463
	PSB3	1.251	0.114	10.925	< .001	1.026	1.475
	PSB4	1.324	0.115	11.535	< .001	1.099	1.549
	PSB5	1.250	0.112	11.133	< .001	1.030	1.470

Factor variances

Factor	Estimate	Std. Error	z-value	p	95% Confidence Interval	
					Lower	Upper
Perceived Security Benefits	1.000	0.000			1.000	1.000

Residual variances

Indicator	Estimate	Std. Error	z-value	p	95% Confidence Interval	
					Lower	Upper
PSB1	0.525	0.090	5.808	< .001	0.348	0.702
PSB2	0.507	0.089	5.706	< .001	0.333	0.681
PSB3	0.485	0.086	5.631	< .001	0.316	0.654
PSB4	0.400	0.078	5.141	< .001	0.248	0.553
PSB5	0.438	0.080	5.480	< .001	0.281	0.595

Average variance extracted

Factor	AVE
Perceived Security Benefits	0.771

Confirmatory Factor Analysis Perceived Usefulness

Model fit

Chi-square test

Model	X ²	df	p
Baseline model	476.411	10	
Factor model	0.411	5	0.995

Note. The estimator is ML.

Additional fit measures

Fit indices

Index	Value
Comparative Fit Index (CFI)	1.000
Tucker-Lewis Index (TLI)	1.020
Bentler-Bonett Non-normed Fit Index (NNFI)	1.020
Bentler-Bonett Normed Fit Index (NFI)	0.999
Parsimony Normed Fit Index (PNFI)	0.500
Bollen's Relative Fit Index (RFI)	0.998
Bollen's Incremental Fit Index (IFI)	1.010
Relative Noncentrality Index (RNI)	1.010

Information criteria

	Value
Log-likelihood	-645.733
Number of free parameters	15.000
Akaike (AIC)	1321.465
Bayesian (BIC)	1360.543
Sample-size adjusted Bayesian (SSABIC)	1313.169

Other fit measures

Metric	Value
Root mean square error of approximation (RMSEA)	0.000
RMSEA 90% CI lower bound	0.000
RMSEA 90% CI upper bound	0.000
RMSEA p-value	0.997
Standardized root mean square residual (SRMR)	0.003
Hoelter's critical N ($\alpha = .05$)	2697.479
Hoelter's critical N ($\alpha = .01$)	3675.615
Goodness of fit index (GFI)	1.000
McDonald fit index (MFI)	1.023
Expected cross validation index (ECVI)	0.304

Kaiser-Meyer-Olkin (KMO) test

Indicator	MSA
PU1	0.901
PU2	0.907
PU3	0.911
PU4	0.926
PU5	0.940
Overall	0.916

Bartlett's test of sphericity

X ²	df	p
459.737	10	< .001

R-Squared

	R ²
PU1	0.829
PU2	0.818
PU3	0.807
PU4	0.762
PU5	0.701

Parameter estimates

Factor loadings

Factor	Indicator	Estimate	Std. Error	z-value	p	95% Confidence Interval	
						Lower	Upper
Perceived Usefulness	PU1	1.334	0.113	11.754	< .001	1.111	1.556
	PU2	1.237	0.106	11.614	< .001	1.028	1.445
	PU3	1.254	0.109	11.485	< .001	1.040	1.468
	PU4	1.249	0.114	10.947	< .001	1.025	1.472
	PU5	1.196	0.117	10.236	< .001	0.967	1.425

Factor variances

Factor	Estimate	Std. Error	z-value	p	95% Confidence Interval	
					Lower	Upper
Perceived Usefulness	1.000	0.000			1.000	1.000

Residual variances

Indicator	Estimate	Std. Error	z-value	p	95% Confidence Interval	
					Lower	Upper
PU1	0.366	0.070	5.213	< .001	0.229	0.504
PU2	0.341	0.064	5.344	< .001	0.216	0.466
PU3	0.376	0.069	5.463	< .001	0.241	0.511
PU4	0.487	0.083	5.854	< .001	0.324	0.649
PU5	0.610	0.099	6.183	< .001	0.417	0.803

Average variance extracted

Factor	AVE
Perceived Usefulness	0.783

Confirmatory Factor Analysis for Acceptance

Model fit

Chi-square test

Model	X ²	df	p
Baseline model	427.031	10	
Factor model	15.278	5	0.009

Note. The estimator is ML.

Kaiser-Meyer-Olkin (KMO) test

Indicator	MSA
ACC1	0.851
ACC2	0.860
ACC3	0.902
ACC4	0.889
ACC5	0.876
Overall	0.875

Bartlett's test of sphericity

X ²	df	p
412.085	10	< .001

R-Squared

	R ²
ACC1	0.800
ACC2	0.742
ACC3	0.723
ACC4	0.741
ACC5	0.726

Parameter estimates

Factor loadings

Factor	Indicator	Estimate	Std. Error	z-value	p	95% Confidence Interval	
						Lower	Upper
Factor 1	ACC1	1.279	0.113	11.302	< .001	1.057	1.501
	ACC2	1.231	0.116	10.614	< .001	1.003	1.458
	ACC3	1.221	0.117	10.420	< .001	0.991	1.450
	ACC4	1.195	0.112	10.634	< .001	0.975	1.415
	ACC5	1.234	0.118	10.442	< .001	1.003	1.466

Factor variances

Factor	Estimate	Std. Error	z-value	p	95% Confidence Interval	
					Lower	Upper
Factor 1	1.000	0.000			1.000	1.000

Residual variances

Indicator	Estimate	Std. Error	z-value	p	95% Confidence Interval	
					Lower	Upper
ACC1	0.410	0.083	4.969	< .001	0.248	0.572
ACC2	0.526	0.096	5.462	< .001	0.337	0.714
ACC3	0.572	0.099	5.770	< .001	0.378	0.767
ACC4	0.500	0.088	5.664	< .001	0.327	0.673
ACC5	0.576	0.101	5.686	< .001	0.377	0.775

Average variance extracted

Factor	AVE
Factor 1	0.746

Descriptive Statistics

Descriptive Statistics

	Mean	Std. Deviation
SA	3.000	1.332
PEU	3.008	1.313
PSB	3.102	1.303
PU	3.034	1.295
ACC	3.080	1.280

Correlation

Pearson's Correlations

Variable		SA	PEU	PSB	PU	ACC
1. SA	Pearson's r	—				
	p-value	—				
2. PEU	Pearson's r	0.992	—			
	p-value	< .001	—			
3. PSB	Pearson's r	0.931	0.928	—		
	p-value	< .001	< .001	—		
4. PU	Pearson's r	0.983	0.983	0.954	—	
	p-value	< .001	< .001	< .001	—	
5. ACC	Pearson's r	0.943	0.936	0.974	0.964	—
	p-value	< .001	< .001	< .001	< .001	—

Mediation Analysis

Parameter estimates

Direct effects

							95% Confidence Interval	
			Estimate	Std. Error	z-value	p	Lower	Upper
SA	→	ACC	0.225	0.151	1.492	0.136	-0.071	0.521
PEU	→	ACC	-0.335	0.154	-2.177	0.029	-0.636	-0.033
PSB	→	ACC	0.575	0.064	9.029	< .001	0.451	0.700

Note. Delta method standard errors, normal theory confidence intervals, ML estimator.

Indirect effects

									95% Confidence Interval	
				Estimate	Std. Error	z-value	p		Lower	Upper
SA	→	PU	→	ACC	0.154	0.068	2.268	0.023	0.021	0.288
PEU	→	PU	→	ACC	0.206	0.077	2.682	0.007	0.056	0.357
PSB	→	PU	→	ACC	0.140	0.042	3.319	< .001	0.057	0.223

Note. Delta method standard errors, normal theory confidence intervals, ML estimator.

Total effects

							95% Confidence Interval	
			Estimate	Std. Error	z-value	p	Lower	Upper
SA	→	ACC	0.379	0.155	2.452	0.014	0.076	0.683
PEU	→	ACC	-0.128	0.153	-0.839	0.402	-0.428	0.171
PSB	→	ACC	0.716	0.055	13.037	< .001	0.608	0.823

Note. Delta method standard errors, normal theory confidence intervals, ML estimator.

Path coefficients

							95% Confidence Interval	
			Estimate	Std. Error	z-value	p	Lower	Upper
PU	→	ACC	0.506	0.136	3.725	< .001	0.240	0.772
SA	→	ACC	0.225	0.151	1.492	0.136	-0.071	0.521
PEU	→	ACC	-0.335	0.154	-2.177	0.029	-0.636	-0.033
PSB	→	ACC	0.575	0.064	9.029	< .001	0.451	0.700
SA	→	PU	0.305	0.107	2.859	0.004	0.096	0.514
PEU	→	PU	0.408	0.105	3.864	< .001	0.201	0.614
PSB	→	PU	0.277	0.038	7.315	< .001	0.203	0.351

Note. Delta method standard errors, normal theory confidence intervals, ML estimator.

R-Squared

	R²
ACC	0.964
PU	0.981

Path plot



